Today's Agenda

Building univariate visualizations in R

- Save your notes in `Getting_Started.Rmd`
- Section: # Visualizing Data in R

ggplot() for making visualizations

Note: You have to load the tidyverse to use ggplot

```
library(tidyverse)
```

Healy (2021) Introduces ggplot()

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = data source, aes(x = variable name))

## 3. Variable type determines geom (choose one)
p + geom_bar()
p + geom_col()
p + geom_histogram()
p + geom_boxplot()
p + geom_line()
```

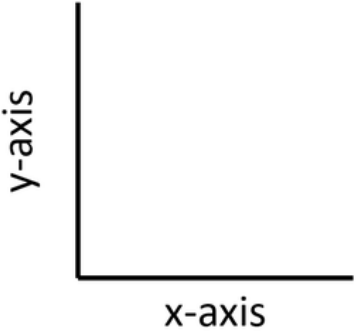
This code is on Discord now!

The GGplot Function: Two Arguments

```
## 1. Tidy and 2. map your data
p <- ggplot(data = data source, aes(x = variable name))
```

1. Identify the source of your tidy data, and
2. Select your variable(s) using the `aes()` function

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = data source, aes(x = x axis variable))
```



- The `x` here refers to the x axis.
- Rotate the plot by changing `x` to `y`

Healy (2021) Introduces ggplot()

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = data source,
            aes(x = x axis variable, y = y axis variable))

## 3. Variable type determines geom (Chang recipes)
p + geom_bar() ## 3.3
p + geom_col() ## 3.1
p + geom_histogram() ## 6.1
p + geom_boxplot() ## 6.6
p + geom_line() ## 4.1
```

Built-in Dataset: chickwts

An experiment was conducted to measure and compare the effectiveness of various feed supplements on the growth rate of chickens.

weight	feed
179	horsebean
160	horsebean
136	horsebean
227	horsebean
217	horsebean
168	horsebean
108	horsebean
124	horsebean
143	horsebean

Built-in Dataset: chickwts

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weight	feed
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168	horsebean
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How do we visualize the `feed` variable?

Built-in Dataset: chickwts

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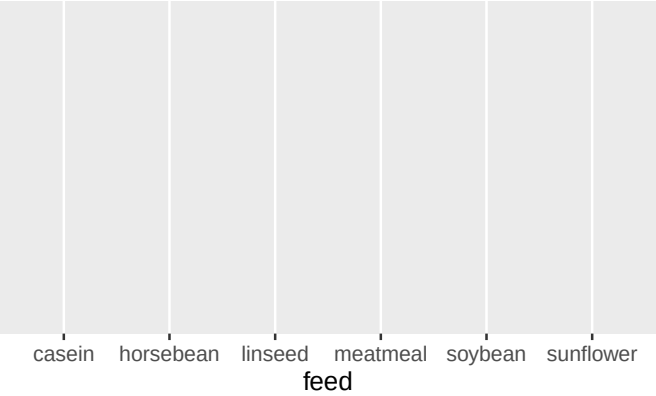
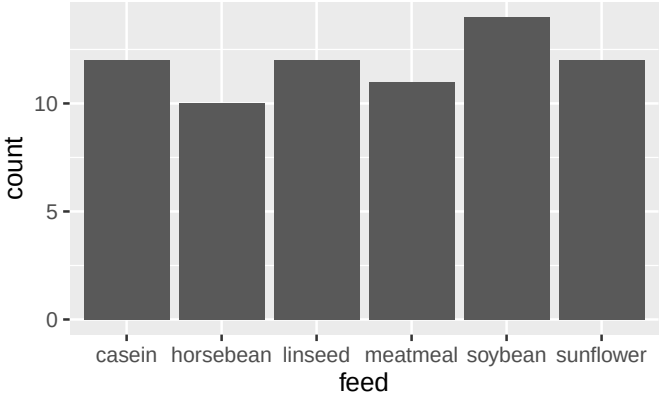
weight	feed
179	horsebean
160	horsebean
136	horsebean
227	horsebean
217	horsebean
168	horsebean
108	horsebean

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = chickwts, aes(x = feed))
```

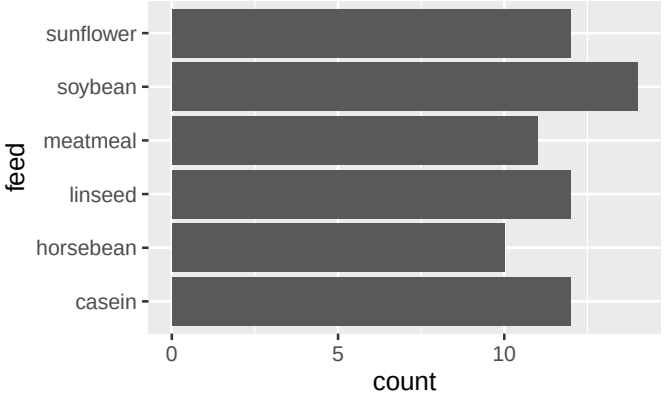
```r
## 1. Tidy and 2. map your data
p <- ggplot(data = chickwts, aes(x = feed))
## View the visualization
p
```



```r
## 1. Tidy and 2. map your data
p <- ggplot(data = chickwts, aes(x = feed))
## View the visualization
p + geom_bar()
```



```r
## 1. Tidy and 2. map your data
p <- ggplot(data = chickwts, aes(y = feed))
## View the visualization
p + geom_bar()
```



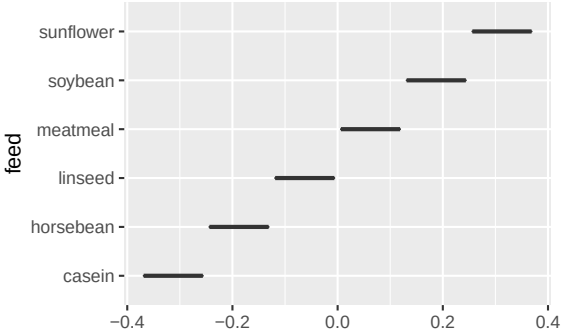
Learn to Decipher the Error Text!

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = chickwts, aes(x = feed))
## View the visualization
p + geom_histogram()
```

Error: StatBin requires a continuous x variable: the x variable is discrete. Perhaps you want stat="count"?

Know what you are making BEFORE you make it

```
## View the visualization
p + geom_boxplot()
```



Bar Plots: Counts (Chang 3.3)

So, if your variable looks like this (categorical):

feed
horsebean
horsebean
horsebean
horsebean
horsebean

```
## geom_bar counts the levels for you
p + geom_bar()
```

How do we make a bar plot if the levels have already been counted?

```r
## Function to count levels
chickwts |>
   count(feed)
```

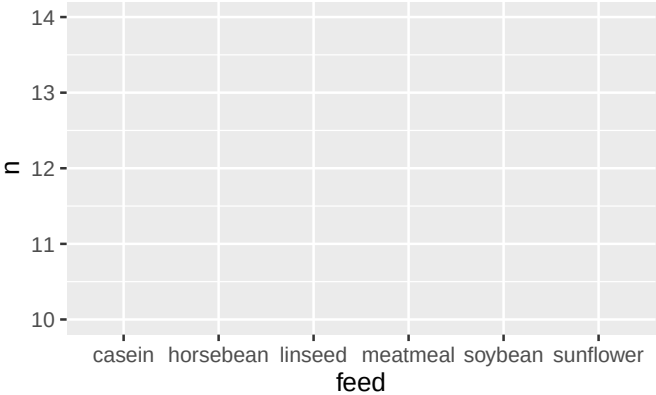
feed	n
casein	12
horsebean	10
linseed	12
meatmeal	11
soybean	14
sunflower	12

```r
## Save the new data
new1 <- chickwts |>
            count(feed)
```

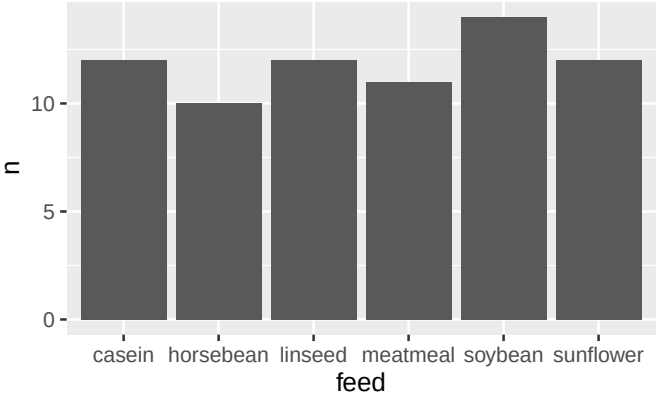
feed	n
casein	12
horsebean	10
linseed	12
meatmeal	11
soybean	14
sunflower	12

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = new1, aes(x = feed, y = n))
```

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = new1, aes(x = feed, y = n))
## View the visualization
p
```



```r
## 1. Tidy and 2. map your data
p <- ggplot(data = new1, aes(x = feed, y = n))
## View the visualization
p + geom_col()
```



geom_bar vs geom_col

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = new1, aes(x = feed, y = n))
## View the visualization
p + geom_bar()
```

Error: stat_count() can only have an x or y aesthetic.

```r
## All in one line (with pipes)
chickwts |>
   count(feed) |>
   ggplot(aes(x = feed, y = n)) +
      geom_col()
```

Built-in Dataset: mtcars

The data was extracted from the 1974 *Motor Trend* US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973-74 models).

	mpg	cyl
Mazda RX4	21.0	6
Mazda RX4 Wag	21.0	6
Datsun 710	22.8	4
Hornet 4 Drive	21.4	6
Hornet Sportabout	18.7	8
Valiant	18.1	6
Duster 360	14.3	8
Merc 240D	24.4	4
Merc 230	22.8	4

Built-in Dataset: mtcars

	mpg	cyl
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Datsun 710	22.8	4
Hornet 4 Drive	21.4	6
Hornet Sportabout	18.7	8
Valiant	18.1	6

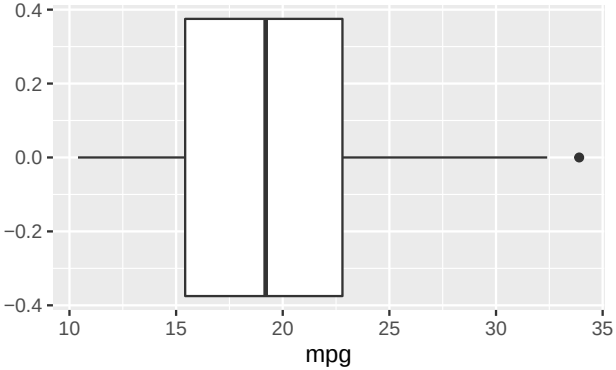
How do we visualize the `mpg` variable?

Built-in Dataset: mtcars

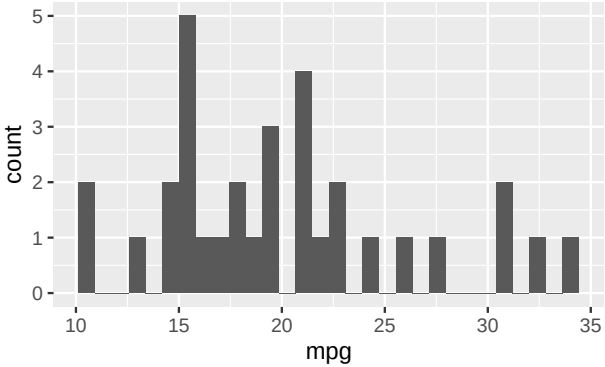
	mpg	cyl
Mazda RX4	21.0	6
Mazda RX4 Wag	21.0	6
Datsun 710	22.8	4
Hornet 4 Drive	21.4	6
Hornet Sportabout	18.7	8
Valiant	18.1	6

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = mtcars, aes(x = mpg))
```

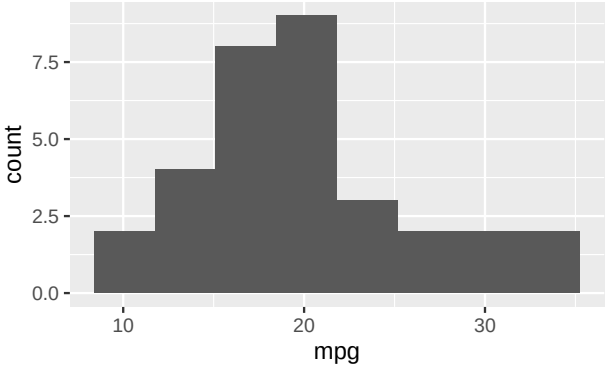
```r
## 1. Tidy and 2. map your data
p <- ggplot(data = mtcars, aes(x = mpg))
## View the visualization
p + geom_boxplot()
```



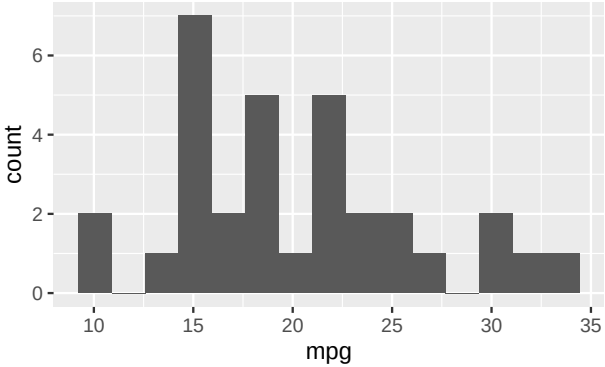
```r
## 1. Tidy and 2. map your data
p <- ggplot(data = mtcars, aes(x = mpg))
## View the visualization
p + geom_histogram()
```



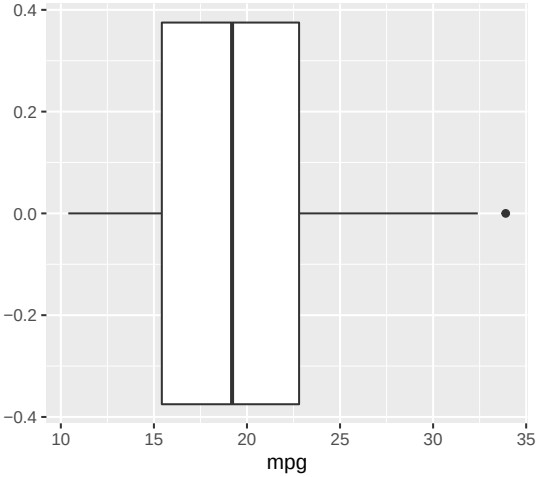
```r
## 1. Tidy and 2. map your data
p <- ggplot(data = mtcars, aes(x = mpg))
## Adjust bin numbers
p + geom_histogram(bins = 8)
```

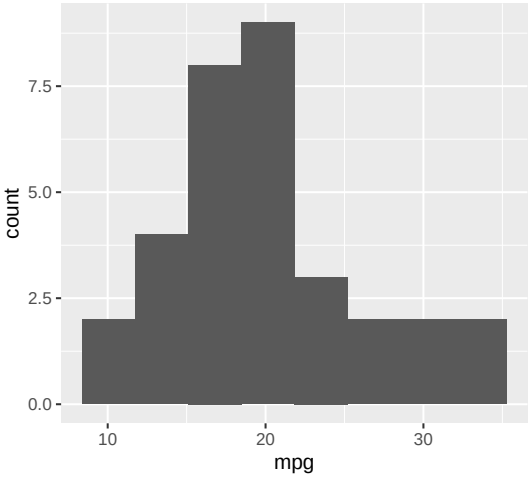


```r
## 1. Tidy and 2. map your data
p <- ggplot(data = mtcars, aes(x = mpg))
## Adjust bin numbers
p + geom_histogram(bins = 15)
```



Comparing Histograms to Box Plots





R Built-in Dataset: economics

This dataset was produced from US economic time series data available from https://fred.stlouisfed.org/.

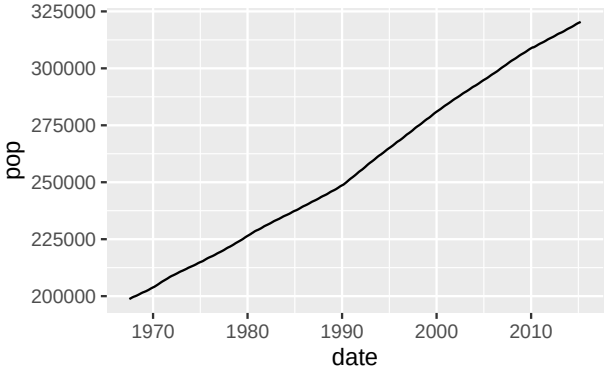
date	pce	pop	psavert	uempmed	unemploy
1967-07-01	506.7	198712	12.6	4.5	2944
1967-08-01	509.8	198911	12.6	4.7	2945
1967-09-01	515.6	199113	11.9	4.6	2958
1967-10-01	512.2	199311	12.9	4.9	3143
1967-11-01	517.4	199498	12.8	4.7	3066
1967-12-01	525.1	199657	11.8	4.8	3018
1968-01-01	530.9	199808	11.7	5.1	2878
1968-02-01	533.6	199920	12.3	4.5	3001

Line Plots (Chang 4.1)

date	pop
1967-07-01	198712
1967-08-01	198911
1967-09-01	199113
1967-10-01	199311
1967-11-01	199498

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = economics, aes(x = date, y = pop))
```

```r
## 1. Tidy and 2. map your data
p <- ggplot(data = economics, aes(x = date, y = pop))
## Line plot (Chang 4.1)
p + geom_line()
```



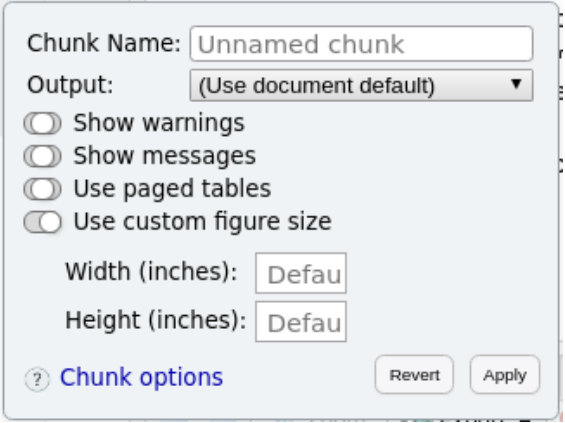
Re-make the visualizations from Tuesday

1. Bar Plots: 1. `CL Rate` and 2. `PR Rate`
2. Box Plots: 1. `CL`, 2. `PR`, 3. `Total`
3. Histogram of the `Total` score
4. Line plot of the **average** `Total` score per year

Controlling Figures in Reports

```{r}
p <- ggplot(data = economics,
            aes(x = date,
                y = pop))

p + geom_line()
```



Analyze a Data Project: Regime Type

Prepare a report **introducing and analyzing** our first data project. What are the **strengths and weaknesses** of this project for measuring regime type?

- Support all claims with evidence.